**CIT 253**
**Assignment 05 – Database Design; Advanced SQL and MySQL**

In the previous assignment, we worked on Chapters 4 and 5 in the textbook. In this assignment, we will go further with MySQL and SQL as described in Chapters 6 and 7 of the textbook.

As the author notes in Appendix A of the textbook, when you installed XAMPP, the database product that was actually installed was MariaDB, which is based on MySQL, and you should have no trouble applying all the MySQL steps shown in the textbook's chapters.

**Part 1:**

To start, do the following steps:

1. Study pages 168 – 188 from Chapter 6 of the textbook – this will mostly be a review of topics you have learned in your Database Design Fundamentals class.
2. Start up the MySQL server – Open the XAMPP Control Panel and click Start next to MySQL to start up the MySQL server.
3. Use your `runClient.bat` batch file from the previous assignment to run the `mysql` client and connect to the server as `root` (or whichever username you set up in the previous assignment). Here were the previous assignment's instructions for this, if you do not remember how to do this step.
    a. Open the command prompt
    b. Change to the folder where you saved the `runClient.bat` file
    c. Type `runClient root` (or type your other username instead of `root`)
    d. You should now be prompted for the password. Enter the password that you set for the root or other user.

**Part 2:**

In this second part of the assignment, you will create two databases, create tables in them, add data to the tables, and execute several SQL queries, all using `SQL` commands via the `mysql` client, as shown in Chapters 6 and 7.

1. Read through both chapters – in Chapter 6, the database and table creation steps begin on page 188.
2. Prepare `.sql` files that contain all of the `SQL` code for creating tables and adding data to them in the two databases that you are asked to create in Chapter 6 – creation of the `forums` database begins on page 188, and of the `banking` database begins on page 198.
3. When you are ready to begin actually creating the databases and the tables inside them, and adding data to them, get a logfile started by using the `tee` command in MySQL. At the command prompt in the `mysql` client, you would type a command in the format

    `tee` *`path to your logfile`*

    So, for this assignment, if you had a folder on your machine at the path `C:\CIT253\Assignment05`, for example, you would type

```
tee C:\CIT253\Assignment05\chapter6_forums.txt
```

and this would result in a file named `chapter6_forums.txt` being created in that folder, containing a log of commands that you type in the `mysql` client and their results.

4. Now set the forums database as the one currently being operated on, with the `USE` command as you did in the previous assignment, i.e.,

```
USE forums;
```

5. Use the `.sql` file or files that you created for the forums database to
    a. Create the tables in it
    b. Add data to the tables
6. Execute the `SQL` queries given for the `forums` database.
7. Stop the logging process by typing `notee` at the command prompt.
8. Exit the `forums` database.
9. Use the `tee` command to start up a separate logfile named `chapter6_banking.txt`.
10. Create the tables and add data using the `.sql` files for the `banking` database.
11. Execute the `SQL` queries given for the `banking` database.
12. Stop the logging process.
13. Now execute the SQL commands from Chapter 7, this time, logging your work in files named `chapter7_forums.txt` and `chapter7_banking.txt`.

Zip all the logfiles along with the `.sql` files you have created, and submit your zip file on Blackboard.